

JUN WEI WANG

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EDUCATION

Fudan University , Shanghai, China	2022 - <i>present</i>
• Software Engineering • Interdisciplinary in Integrated Circuits (集成电路) • Chinese Government Scholarship, CSC Scholarship Category A	
Tongji University , Shanghai, China	2021 - 2022
• Pre-university Chinese (预科) • CSC Scholarship Category A • Top scoring CSC pre-university graduate	
Elgin Park Secondary , Surrey, British Columbia, Canada	2018 - 2021
Grades 10 - 12	
• Tiger Sun Memorial Computer Science Scholarship ▸ Exceptional computer science student & entering university to study computer-related courses • 2020/21 BC Achievement Scholarship ▸ One of 8,000 graduates with top academic achievement in British Columbia	
Nisga'a Secondary School , New Aiyansh, Canada	2017 - 2018
Grade 9	
Elgin Park Secondary , Surrey, British Columbia, Canada	2016 - 2017
Grade 8	

TECHNICAL SKILLS

- **Programming:** Python; C/C++ (Qt; Basic OpenGL; drivers/system-level; embedded on Arduino/ESP32; CUDA Platform); JavaScript (TypeScript); Java; Rust
- **Full-stack developer:** End-to-end web apps; integration across frontend, backend, APIs, and data stores; familiar with customizing JS compilers (Webpack, SWC, Rollup, etc.); SPA/SSR/ISR solutions;
 - **Frontend:** React/Next.js; TypeScript; TailwindCSS; common UI libraries
 - **Backend:** Node.js with Express; REST/GraphQL APIs; PostgreSQL/MySQL; Redis; common backend frameworks
- **Cloud & DevOps:** AWS (EC2/S3; static/backend hosting); Aliyun (ECS/OSS; hosting); GitHub CI/CD basics

PROJECT EXPERIENCE

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|--|-----------------------|
| Compass HLS DSE Application | 2025 - <i>present</i> |
| PI — FDUROP (Fudan Undergraduate Research Opportunities Program) | |
| <ul style="list-style-type: none">• GUI for Compass HGBO-DSE (Hierarchical GNN + Bayesian Opt) to auto-generate HLS pragmas• Led VS Code plugin + Svelte front-end; integrated AST-based pragma synthesis and Vitis HLS backend• Custom server backend | |

- Custom Python inference backend
- Future work includes integration of a newer model and optimizations to PPA prediction depending on FPGA architecture

Sichain Semiconductor

2025 - *present*

MOSFET manufacturing — testing & QC pipeline

- ML-aided AOI (ResNet-18/50) for wafer/assembly defect detection
- Rust + React (Tauri) tooling (with ThreeJS for rendering) to ingest and streamline QC stages across fabs/assembly (including CP, WLBI, AOI, etc.)

Jarvis (E-bike CMS)

2025 - *present*

ESP32S3 BLE/Wi-Fi/4G central management/monitor system

FDE-CLI

2025

Rust CLI tool for FDP3P7 (SMIMS) board (Fudan's in-house FPGA board for FPGA Principles & Applications course)

- Drives SMIMS engine: program bitstreams, R/W registers & FIFOs; multi-threaded CLI with debugging helpers; coursework artifacts in fde-project
- Master/multi-slave BLE, SoftAP + web UI, LTE/IoT telemetry, GPS tracking, anti-theft/unlock; integrates Far-driver ESC + AntBMS; ESP-IDF + Svelte framework embedded front-end

DNN Acceleration on FPGA

2025

Hand-written number classification (MNIST) CNN on FPGA (PL) with Linux PS + Ethernet I/O

- Implemented a CNN in PL logic; PS side bridged to PL and exposed Ethernet for data in/out between FPGA and external devices
- Built a custom Python client over Ethernet; delivered live webcam handwritten-digit recognition pipeline;

Loong Squad

2024 - 2025

Superscalar OoO (out-of-order) LoongArch32 CPU (Chisel)

- LoongCore: superscalar OoO CPU (LoongArch32) with branch prediction, caches, pipeline flush handling; Scala/Chisel + SBT
- Team prize: 2nd Place, Loongson Cup group contest (Loong Squad@FDUPPI);
WeChat post: WeChat
- Results: prelim 1st place, finals 6th overall; ~5.3x speedup on provided tests at ~100 MHz
- Passed FuncTest/PerfTest/SysTest and boots Linux on LoongSoC

Full-stack Web Platforms

2024 - *present*

Next.js + DO + MySQL + Cloudflare + S3

- Built and hosted multiple in-progress apps with custom domains: glorialan.com (Next.js), backend, and jimmys-cms
- DigitalOcean hosting with managed MySQL; domains via GoDaddy; Cloudflare DNS/CDN; static assets on Amazon S3

TeaPl Compiler

2024

Course project: Custom compiler (built with C++)

- Built front to back: lexer/parser, AST, IR, optimizations, ARM assembly codegen, verification for teaching language **TeaPl**
- TeaPL language spec: teapl.md

	DSP BiTCN Post-EQ 2024 ML post-equalization for 64QAM-CAP • BiTCN-based post-equalizer for 64QAM-CAP signals; references bidirectional TCN-based nonlinear compensation in underwater VLC; midterm DSP project
	LoongArch32 Verilator Debugger 2023 GDB-like Verilog CPU debugger (C++/Verilator) • Built a gdb-style CLI debugger (fdb) for LoongArch32 labs using Verilator & C++; supports step/run/break/print on bin images • Make/IMG-driven flow for swapping binaries
	Single-Cycle MIPS CPU 2023 Course project: Educational RISC pipeline • Implemented single-cycle MIPS CPU for coursework/labs • Supported instructions: add, sub, and, or, slt, addi, andi, ori, slti, sw, lw, j, nop
WORK EXPERIENCE	Fysics / 飞捷科思 (fysics.net) 2024-09 - <i>present</i> Intern — SDF algorithms, CUDA acceleration, & software development • In-house physics simulation engine with an OpenGL + ImGui frontend • SDF algorithms and related pipeline improvements • GPU acceleration with CUDA • Software development across data, algorithms, and app integration English Tutor 2023-09 - <i>present</i> Private instructor • 1:1 tutoring focused on English reading, writing, and conversational fluency • Designed individualized study plans and practice materials Freelance (Fiverr) 2021 - 2022 Frontend & CS projects; tutoring • Built React/TS frontends and small apps; custom dashboards and UI refactors • Tutored CS topics (data structures/algorithms) and tech interview prep; project support
ACHIEVEMENTS	• 2025 National College Student Computer Systems Ability Contest — CPU Design (Loongson Cup) ▸ Team name: Loong Squad@FDUPPI - Second Place • Mathematical Content in Modeling (MCM) - 2025 ▸ MCM Problem C (data insights) ▸ Honorable Mention • 2024 National College Student Computer Systems Ability Contest — CPU Design (Loongson Cup) ▸ Individual competition • Fudan University 4th Classic English Speech & Reading Contest — Second Place — 2024.05.18 • Advanced Placements (AP) ▸ Calculus BC (5/5) ▸ AP Computer Science A (5/5) • Canadian Computing Competition University of Waterloo

	▸ 2020 Senior ▸ 2019 Junior	
SOFTWARE COPYRIGHTS	• AI-assisted High-Level Synthesis Optimization GUI Tool (Compass) V1.0; Reg. No. 2025SR2155133; Certificate No. 软著登字第 16811331 号; Rightsholder: Fudan University; Granted 2025-11-05	
ACTIVITY EXPERIENCE	HGBO-DSE demo @ FPGA Education Workshop 2025-07 Co-presenter with advisor • Presented “HGBO-DSE: Hierarchical GNN and Bayesian Optimization based HLS Design Space Exploration” alongside faculty at the 大模型赋能 FPGA 教育研讨会 • Demoed the HGBO-DSE algorithm and the Compass GUI (HLS DSE) Elgin Park Secondary 2019 - 2021 Programming Club (founder) & 3D Printing Club (co-founder) • Launched the school’s first programming club (Gr. 11-12); taught C++/Java; prepared peers for Waterloo CCC Junior; weekly labs filled the CS room • Led 3D printing club: modeled parts, ran on-site printing service, trained students on 3D workflow, printers, and troubleshooting	
HOBBIES AND INTERESTS	• Violin: ~9 years ▸ The Royal Conservatory of Music (RCM): Level 4, level 8 ▸ Vancouver Academy of Music (VAM) – VAMIS (Intermediate Symphony; first violin) – VAMSO (Symphony Orchestra; performing major symphonic works at a pre-professional level; second violin) ▸ Surrey Youth Orchestra (second violin) • Photography: Street/travel and landscape; exhibitions ▸ Tongji University 2022 International Student Photography Exhibition (作品展) • Hardware & IoT tinkering: simple circuit design & soldering; Arduino projects; Raspberry Pi projects; ESP32 wireless smart home modules; Home Assistant; custom 433 MHz Arduino smart home nodes (MySensors); home server/networking; etc. • Full-stack web dev: multiple web apps; React/Vue/Next; Firebase/MongoDB/DigitalOcean/AWS; self-hosted servers and own my own domain name • Programming, reverse engineering, hacking: love to learn about low-level tinkering, tooling, and security/vulnerability related topics	

